Universitat Rovira i Virgili (URV) y Universitat Oberta de Catalunya (UOC)

Master in Computational and Mathematical Engineering

FINAL MASTER PROJECT

Area: YYY

xxx title of the work xxx

xxx subtitle (if it is the case) xxx

Autor: Complete name of the student

Tutor: Complete name of the director

Barcelona, November 17, 2025

Dr./Dra. (name) ...., certifies that the student (name) .... has elaborated the work under his/her direction and he/she authorizes the presentation of this memory for its evaluation.

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A page with the specification of credits / copyright for the project (either application on one hand and documentation on the other, or unified), as well as the use of brands, products or services of third parties (including source codes). If a person other than the author collaborated in the project, his identity must be made explicit and what he did.

The most usual case is exemplified below, although it can be modified by any other alternative:

FINAL PROJECT SHEET

Title:	Descriptive of the project
Autor:	Name and surname(s)
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Date (mm/yyyy):	Date of delivery
Program:	Master in Computational and Mathematical Engineering
Area:	Name of the FMP area
Language:	English
Key words	Maxim 3 key words

Dedicatory / Quote

Brief words of dedicatory or quote.

Acknowledgments

If deemed appropriate, mention the persons, companies or institutions that have contributed to the realization of this project.

Abstract

Text with the synthesis of the project, that is, a text in which the definition of the project / problem addressed, its objectives / methods of resolution, and the results and conclusions are briefly explained (it can not be a list, but a text continuous written in a structured way). If it is necessary to put a reference in this text, it will be annotated at the bottom of the same page. In this section you can use a more literary and colloquial language than for the rest of the document.

In case of not writing the the document in English, it will be necessary to write the Abstract in duplicate. One of the versions must be textbf obligatorily in English. The other version can be written in Catalan or Spanish, in the language used for the rest of the report. The word Abstract will be changed to " textbf Resum " or " textbf Resumen " in the Catalan and Spanish versions, respectively.

Recommended length: 250 words maximum.

How to write a good abstract (in English):

http://www.ece.cmu.edu/ koopman/essays/abstract.html

Keywords: Work keywords separated by commas. For example for this document they could be Model, Guideline, Template, Memory, Final Master Project / Master

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Chapter 1

Introducción

1.1 General description of the problem

At present, data mining processes require large amounts of data, which in many cases contain personal and private information of users or people. Although basic anonymization processes are carried out on the data, that is, elimination of names or other key identifiers, there are a multitude of re-identification techniques that allow to re-identify a user within this data set. Figure ref fig: context-anoni1 presents a map where it is possible to contextualize the processes of anonymization and re-identification within a process of data mining.

1.1.1 Example of subsection

Although important advances have been made in the preservation of privacy in the publication of data, such as the model k-anonymity [1].

An example of a pseudo-code can be found in the Code 1

An example of a table can be seen in the Table 1.1

Node ID	$\mathcal{H}_0$	$\mathcal{H}_1$	$\mathcal{H}_2$
Alice	ϵ	1	$\{4\}$
Bob	ϵ	4	$\{1, 1, 4, 4\}$
Carol	ϵ	1	$\{4\}$
Dave	ϵ	4	$\{2, 4, 4, 4\}$
Ed	ϵ	4	$\{2, 4, 4, 4\}$
Fred	ϵ	2	$\{4, 4\}$
Greg	ϵ	4	$\{2, 2, 4, 4\}$
Harry	ϵ	2	$\{4, 4\}$

Table 1.1: *Vertex refinement queries.*

Figure 1.1: Foot of the image.

Algorithm 1 Pseudocode of the algorithm *Random Switch*

Entrada: The original graph G and the percentage of anonymization p that you want to apply.

Salida: El graph G anonymized

$num = round(G.num\_edges() * p)$
$i = 0$
while $i < num$ **do**
  $e_1 = G.random\_edge()$
  $e_2 = G.random\_edge()$
  $new\_e_1 = (e_1.origen, e_2.origen)$
  $new\_e_2 = (e_1.destino, e_2.destino)$
  if $!G.exist(new\_e_1)$ **and** $!G.exist(new\_e_2)$ **then**
    $G.add\_edge(new\_e_1)$
    $G.add\_edge(new\_e_2)$
    $G.delete\_edge(e_1)$
    $G.delete\_edge(e_2)$
    $i = i + 1$
  end if
end while
return G

Bibliography

[1] Latanya Sweeney. k-anonymity: a model for protecting privacy. *Int. J. Uncertain. Fuzziness Knowl.-Based Syst.*, 10:557–570, October 2002.